A BAGGAGE DELAY IS NOT JUST A BAGGAGE PROBLEM

During disruptions, baggage, flight and ground-resource signals sit in different systems and teams — so the **bottleneck, its impact and the next action are found late.**

REAL INCIDENT · SUVARNABHUMI, 25–30 SEP 2026

Heavy rain (116 mm on 25 Sep) → **Flooded roads** (CAAT notice, 26 Sep) → **Staff can't arrive** (baggage carts failed too) → **Baggage backlog** (10,000+ bags in 4 days) → **Flights affected** (cuts · cargo suspended)

On 29 Sep Thai Airways' CEO said ~5,000 bags remained; ~8,000 were still outstanding on 30 Sep. 131 military personnel were deployed, cargo acceptance was suspended 30 Sep–6 Oct and flights were reduced.

24M
bags mishandled worldwide in 2025
SITA 2026

$6.3B
annual industry cost (~$260 per bag)
SITA 2026

39%
of mishandling happens at transfer
SITA 2026

The question we chose: when baggage is delayed, where is it stuck, why — and what do we do first?

Sources: Open-Meteo (rain), CAAT 30/2569, The Nation, The Star (incident), SITA Baggage IT Insights 2026. “Mishandled” = delayed, damaged or misrouted — not permanently lost. News figures change day to day.

WHO FEELS IT — AND WHERE IT BREAKS

PRIMARY USER

Baggage / Ground Operations Controller

Decides every minute where to send people and equipment first — with data spread across systems, and a lot of calling and walking to confirm it.

SECONDARY USERS & BENEFICIARIES

Airline OCC — wait for bags or send them on

Airport Operations — see across teams

Ground handlers — get a work order when short-staffed

Passengers — bags arrive with them

BAGGAGE JOURNEY — EVERY ARROW IS A HANDOFF BETWEEN TEAMS

Check-in → Screening → Sorting → Loading → Transfer → Arrival

SKY itself points to handoffs as where baggage goes wrong, and SITA finds **39%** of mishandling occurs at transfer — many parties, limited connection time.

When baggage stops moving, who knows WHY first?

Sources: SKY Innovation Spark Kick-off (1 Oct 2026) on baggage process and handoffs; SITA Baggage IT Insights 2026.

ONE TRUSTX: BAGGAGE DISRUPTION INTELLIGENCE

Combines baggage, flight, resource and disruption signals to **detect bottlenecks early, predict downstream impact and recommend recovery priorities** — the operator decides.

Data Integration
Scans, flight schedule and staffing become one event format.

Trusted Event History
Every record has a source, a time and a freshness check — auditable.

Common Operational State
One shared status of the six baggage stages.

AI Decision Intelligence
Detect bottleneck · Predict at-risk flights · Recommend order · Coordinate owners

OVERLAY, NOT REPLACEMENT

We don't build another baggage tracker

BRS already tracks bags. We take signals from BRS, AODB and ground teams and work out where it is stuck, what it affects and what to do first.

FOUR QUESTIONS, ONE SCREEN

① Is it normal?

② Which stage is stuck?

③ Which flights, how many bags?

④ What first — and by when?

! Same four layers as One TrustX in round one, applied to baggage data. The system recommends; people decide; no airport equipment is controlled.

FROM SIGNAL TO DECISION

Ingest
scans · schedule · staffing

→

Verify & label
source · freshness · duplicates

→

Shared state
6 stages: waiting, how long

→

Detect
stage over threshold

→

Predict
flights that won't make it

→

Recommend
order by hold-close

→

Human decides
approve · reject · edit

EXAMPLE FROM THE MODEL · FLOOD SCENARIO · 08:30

SIMULATED DATA

OK	OK	OK	DELAY	OK	OK
Check-in	**Screening**	**Sorting**	**Loading**	**Transfer**	**Arrival**
entry	0 bags	27 bags · 5 min	202 bags · 35 min	24 bags · 25 min	129 bags · 25 min

IMPACT 3 of 10 flights at risk · 60 bags at risk, 44 transfer

CAUSE (if ground data connected) staff at 55% of plan, carts 4 of 5

RECOMMEND give loading capacity to TT403 first (closes 08:55), then TT404, TT405

OWNER: GROUND HANDLER · HUMAN APPROVAL REQUIRED

Without ground-handler data the system still shows **where** it is stuck — and says “cause unknown” instead of guessing.

! Computed by the simulation (seed 42) on synthetic data — not real airport data. Try it at onetrustx.vercel.app/baggage

HOW WE DIFFER FROM WHAT SKY ALREADY RUNS

SYSTEM	ANSWERS	ONE TRUSTX ADDS
BRS / BHS	Where is this bag? Right aircraft? Alerts for no-show bags and wrong container.	Reads flow by stage against flight deadlines: where it is stuck and which flights are at risk.
AODB	When does each flight depart and arrive, which gate.	Uses it as the deadline for prioritising — no duplication.
A-CDM	Airport partners working to flight milestones.	Not rebuilt. Adds per-flight baggage-readiness risk as one more input.
Ground handler	Knows own staffing, not the whole baggage line.	Gets the order of flights to serve first when people are short.

THE LINE TO REMEMBER

BRS tracks the bag.
One TrustX manages the disruption around the bag.

We build on existing signals — we don't claim BRS can't do something.

TRUSTED DATA

Every event keeps its source, time and the decision made on it — traceable end to end.

! BRS capabilities are taken from SKY ICT's public project page; actual deployments may do more — to be confirmed with SKY. A-CDM and AODB as described at Kick-off.

BUILDABLE — ON DATA SKY ALREADY HAS

DATA NEEDED — AGGREGATE COUNTS, NO PERSONAL DATA

BRS / BHS	Scan counts per stage, per flight	Required
AODB	Departure times, on-block times, gate	Required
DCS / BRS	Expected bags per flight, connecting flight	Required
Ground handler	Staff vs plan, baggage carts	Cause
BHS	Belt and sorter status	Cause

To confirm with SKY: BRS can export per-stage scan timestamps as aggregate counts.

RISK → MITIGATION

No handler data → still works, just no cause

Scans lost or late → asks for verification, never guesses

Thresholds differ in reality → validate with experts first

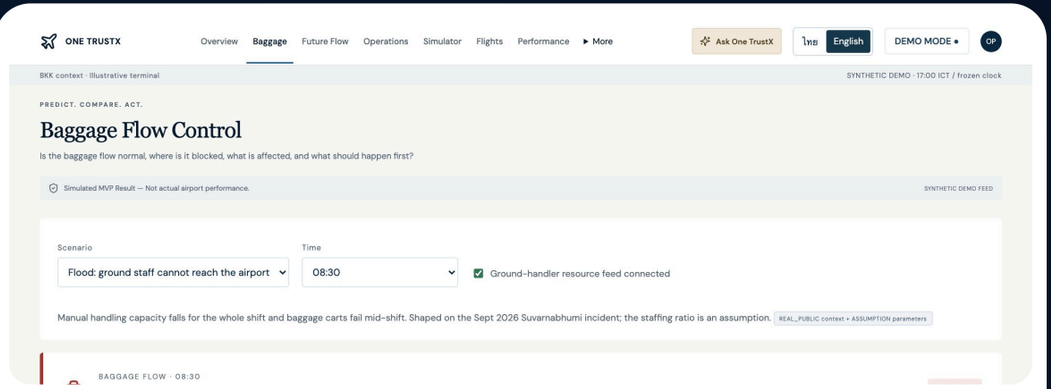
How it would run

① Read-only

② A human approves every recommendation

③ Controls no equipment

④ If One TrustX stops, existing systems carry on



Working prototype: onetrustx.vercel.app/baggage (synthetic data)

! The prototype runs on synthetic data with every parameter labelled (15: real 2 · benchmark 1 · derived 1 · assumption 11). Main risks: access to ground-handler data and expert validation of thresholds.

IMPACT WE EXPECT — AND CAN MEASURE

BAGS LEFT BEHIND · FLOOD SCENARIO · 30 RUNS COMBINED

SIMULATED

Capacity shared equally between flights (today's baseline — an assumption)



9,082

Served in the order One TrustX recommends



4,227

53.5% fewer bags left behind

9,082 → 4,227 bags over 30 runs

62.9% fewer transfer bags left behind

1,862 → 690 bags

87% F1 of flight-level warnings

precision 0.91 · recall 0.84

55 min warning before first bag is missed

range 25–125 min

NOT MEASURABLE YET — we will measure it in the prototype phase

People's coordination time: controllers run the same scenario with and without the screen; time from first signal to an owner accepting the task.

Flight delay minutes: needs a rule for holding flights, or comparison with historical data.

40% → 7% status wrong without a warning, when data has faults

7,804/7,804 injected faults caught · 0/87,890 good records rejected

! 30 simulated runs per scenario on synthetic data — not airport results. The reduction depends on the assumed baseline (equal sharing); if handlers already prioritise well, the gain is smaller. Ground truth comes from the same simulation.

PLAN FOR THE PROTOTYPE PHASE

1

Validate with experts

Thresholds, hold-close time, how handlers prioritise today, whether BRS exports per-stage data.

2

Refine scenarios

Add inbound-bag backlog as in the real incident; Normal · Staff shortage · Sorter fault · Flood.

3

Test with users

Controllers use the screen; measure coordination time, false alerts and usefulness of advice.

4

Shadow mode

Anonymised historical data, read-only, compared with what actually happened.

VISION — EXPAND ONLY AFTER THE VALUE IS PROVEN

Baggage → Turnaround → Gate → Passenger → Cargo

Start with baggage, where every minute and every handoff matters.
Then grow into a trusted data layer for the Future Aviation Hub.

! Dates other than the 22 Oct 2026 pitch will follow the SKY Innovation Spark organisers' announcements.